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Unraveling multiparty conversations: From human interaction mechanisms to conversational agent challenges and persona design

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HIGHLIGHTS

- An empirical analysis of 5 key mechanisms that underpin successful human-human text-based multiparty conversations.
- An identification of 4 traits that constitute an ideal conversational agent persona for multiparty settings.
- A conceptual model synthesizing these mechanisms and persona traits for designing CAs in multiparty participation.

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ABSTRACT

Multiparty conversations are ubiquitous and indispensable in diverse social and collaborative contexts. However, current conversational agents (CAs) face significant challenges in effectively engaging in such interactions, particularly within text-based environments. While earlier limitations were often attributed to the inadequacies of AI models, recent advances in large language models now compel us to revisit both our understanding of multiparty conversation and the way we design CAs. This paper synthesizes findings from two complementary qualitative investigations and proposes a conceptual model for designing CAs that can genuinely participate, rather than merely function as tools or outsiders. The first study, employing retrospective think-aloud sessions (N = 30) with users in text-based multiparty settings, uncovers 5 key interactional mechanisms (e.g., Turn-taking Management, Presence Management) that underpin successful human-human multiparty interactions, derived from participants' articulated perceptions and reasoning. Subsequently, the second study, through semi-structured interviews (N = 15), identifies user expectations for CA integration and key traits (e.g., proactivity, social authenticity) that shape an ideal CA persona perceived by users as a genuine participant. Drawing from these human-centric insights, we then derive design considerations, aiming to guide the development of CAs capable of more natural, effective, and socially intelligent participation in multiparty conversation.

1. Introduction

Multiparty conversations are a central part of human interaction, allowing more than two participants to exchange information, coordinate activities, and build relationships in both social and collaborative settings (Miller, 2007). Given their ubiquity, understanding the distinct dynamics in multiparty conversations has been a long-standing focus in the field of human-computer interaction (HCI) and computer-supported cooperative work (CSCW). By modeling multiparty conversations across diverse domains such as education (Thornborrow, 2003; Stokoe, 2000), healthcare (McCabe et al., 2002), workspace (Handel and Herbsleb, 2002), and entertainment (Morrison and Martens, 2017), researchers have gained valuable insights into promoting effective communication, enhancing collaboration, and fostering relationship-building.

Concurrently, advancements in artificial intelligence (AI) have fueled growing interest in extending the capabilities of conversational agents (CAs) beyond dyadic interactions to include group or multiparty contexts (Seering et al., 2019). However, the inherent complexity of multiparty conversations presents significant challenges for CAs to engage naturally. The fluidity of topic shifts, the coexistence of parallel conversation threads, and the need to manage social dynamics remain

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difficult for AI systems to navigate in a human-like manner, often limiting CAs to the role of a tool or external coordinator rather than a genuine participant (Shamekhi, 2020). While data-driven approaches, such as training natural language processing (NLP) models on discourse structures or turn-taking prediction (Afantenos et al., 2015; de Bayser et al., 2019), have shown promise, their success has largely been limited to simple, task-oriented CAs (Avula et al., 2018; Kim et al., 2020, 2021). With the remarkable advancements in Large Language Models (LLMs), the challenge of enabling CAs to genuinely participate in multiparty conversations has shifted from a purely technical problem to a more fundamental design problem.

To inform this design-focused approach for multiparty CAs, we conducted two complementary user studies to gather insights about designing CAs for multiparty conversation. In the first study, we revisit human-human interaction in text-based multiparty settings to better understand the mechanisms that allow people to manage these complexities through a **retrospective think-aloud approach**. Instead of relying solely on a descriptive analysis of micro-level interactions of conversation history through conversational analysis, this study gathers insights directly from participants' perceptions and reasoning processes, by prompting participants to verbalize their thoughts after engaging in conversations. The second study employed **semi-structured interviews** to explore how users conceptualize CAs in multiparty conversations. We then applied thematic analysis across both studies to systematically analyze and interpret the collected data.

In the first study, we exploratorily identified 5 essential mechanisms, drawing from participants' articulated cognitive and social reasoning, that enable humans to successfully navigate and genuinely participate in text-based multiparty conversations:(1) Turn-taking Management, ensuring structured participation through systematic turn-allocation and prevalent self-selection; (2) Thread Management, involving the initiation, progression, and closure of conversational threads for coherence and engagement; (3) Relevance Management, which maintains cohesion and fosters common ground through purposeful contributions; (4) Presence Management, enabling participants to signal attention and involvement to promote balance and inclusivity; and (5) Role Management, concerning the identification and adaptation of roles for dynamic participation and smooth coordination.

Building on this foundation, our second study explored users' envisioned integration of CAs into multiparty conversations, delving into their expectations for an agent's roles and desired traits, alongside significant user concerns. Synthesizing these user-centric insights, we identified 4 core traits that constitute an ideal CA persona: being proactive, maintaining long-term relationships, exhibiting social authenticity, and displaying human-like imperfections.

Integrating the insights from our two studies, this paper proposes a conceptual model (Fig. 4) that illustrates how CAs can achieve genuine participation. This model serves as the foundation for a set of derived design considerations aimed at developing agents that are capable, believable, acceptable, and socially well-integrated. Furthermore, the paper discusses broader implications arising from these findings, including those related to CA agency and autonomy, and rethinks the concept of CA humanness within the multiparty context.

In summary, this paper presents the following key contributions:

- An empirical analysis of 5 key interactional mechanisms that underpin successful human-human, text-based multiparty conversations.
- An identification of 4 CA traits that constitute an ideal persona, grounded in user expectations for social intelligence and interaction in multiparty settings.
- A conceptual model that synthesizes these mechanisms and persona traits to guide the design of CAs capable of genuine multiparty participation.

2. Related work

2.1. Modeling human conversation for interaction design

Conversation is one of the most natural, frequent, and universal aspects of human interaction (Miller, 2007), allowing two or more people to exchange information, express emotions, build relationships, and coordinate activities (Luppicini, 2008). It is inherently dynamic and context-dependent, characterized not only by a sequence of utterances exchanged between participants, but also by underlying intentions, nonverbal cues, and cultural-societal norms (Gumperz, 1982; Miller, 2007). Therefore, understanding how people actually organize conversation — down to the timing of pauses, overlaps, and topic shifts — has long been argued as a prerequisite for enhancing computer-mediated communication and designing interactive systems that feel natural and trustworthy.

This need to understand the fine-grained structure and organization of talk-in-interaction has given rise to the field of Conversational Analysis, a qualitative research methodology dedicated to describing the orderliness, structure, and sequential properties of conversation (Goodwin and Heritage, 1990; Hutchby and Wooffitt, 2008). In their seminal work, Sacks et al. introduced a foundational model of turn-taking, demonstrating how speakers allocate speakership and manage the transition between turns through a locally organized, finely timed process (Sacks et al., 1978). Grounded in the rigorous empirical inspection of naturally occurring dialogue, researchers have identified additional mechanisms central to conversation, including repair — the practice of addressing and resolving communicative breakdowns (Schegloff et al., 1977) — as well as action formation (Levinson, 2012) and sequencing (Schegloff, 2007), which investigates how participants implement social actions (e.g., requests, assessments, invitations) through talk and how these actions are embedded in and gain meaning from the larger sequential environment. While Conversation Analysis started with a focus on casual conversation, its methods were subsequently adapted to task-oriented interactions. For example, its application in healthcare includes assessing and treating individuals with aphasia (Beeke et al., 2007), while in education, it supports fostering interactional competence among second-language learners (Barraja-Rohan, 2011).

Responding to Norman and Thomas's call to translate Conversation Analysis insights into actionable HCI guidelines that keep interaction sequences aligned with users' social expectations (Norman and Thomas, 1990), Conversation Analysis has more recently been widely adopted to analyze user interactions with CAs — such as voice assistants (Porcheron et al., 2017, 2018; Wu et al., 2022), chatbots (Li et al., 2020), and social robots (Pelikan and Broth, 2016) — thereby informing future designs and enhancing user experience. For example, Li et al. (Li et al., 2020) identified "non-progress" issues and corresponding user coping strategies in interaction with task-oriented chatbot, offering design implications that minimize non-progress and better guide users. Similarly, Wu et al. (Wu et al., 2022) examined how native and non-native speakers construct commands for Google Assistant, proposing design improvements to support more inclusive personal assistants. Porcheron et al. (Porcheron et al., 2017) analyzed voice-assistant use in co-located group settings, revealing that participants treat the agent as a non-human resource, use distinctive repair strategies with it, and rely on visible cues to keep the interaction accountable to everyone present. These studies collectively offer valuable insights for developing the conversational and social intelligence of modern CAs.

However, most Conversation-Analysis-informed CA studies still focus on dyadic interactions — whether traditional human-human pairs, user-agent exchanges, or one-to-one agents embedded in group settings — leaving the nuanced dynamics of genuine multiparty conversation largely unexplored. In this paper, we intend to investigate these multiparty conversation dynamics more deeply to derive design implications for CAs in these settings. A further consideration is that while Conversational Analysis provides rich insights into the observable

organization of talk, it typically describes how interaction unfolds without fully addressing the underlying cognitive processes or motivations that shape why participants organize talk in a particular way (Potter and Edwards, 2012). Drawing on the perspective of Suchman's situated-action theory (Suchman, 1987), which emphasizes how actions are constructed moment-to-moment based on evolving circumstances, we believe that understanding participants' internal reasoning is necessary for a complete, in-situ grasp of how they participate and contribute. Accordingly, we adopted a retrospective think-aloud study (Study 1) to complement the understanding offered by Conversational Analysis and provide a richer empirical basis for informing CA design.

2.2. Handling multiparty conversation

Multiparty conversations involve interactions between three or more participants (Miller, 2007). They are ubiquitous in both social and collaborative settings, occurring countless times during daily activities such as friends gathering, online group chatting, family dinner, work meeting, and more. For a long time, multiparty conversations were overlooked, largely due to the belief that they were merely a simple addition of dyadic conversations (Cooney et al., 2020). Consequently, most existing CAs are designed with a one-on-one interaction paradigm. With the growing demand for collaboration and complex human-AI interactions, there has been a renewed effort to understand and differentiate multiparty conversations from dyadic conversations.

The dynamics of multiparty conversations resemble the "three-body problem" in physics, where interactions among three or more participants introduce exponential complexity. This makes multiparty conversations fundamentally distinct from dyadic interactions and presents significant challenges for analysis (Traum, 2004). In more detailed aspects, Kangasharju and Lerner examined the dynamic formation of teams or collectivities within multiparty conversations, highlighting its impact on participation (Kangasharju, 1996; Lerner, 1993). Aoki et al. identified the presence of multiple simultaneous conversational floors—active conversational spaces or threads—along with their initiation and how participants signal affiliation with them (Aoki et al., 2006). Cooney et al. further emphasized the "many minds problem", where reduced airtime per participant, complex turn-taking, and limited listener feedback make self-disclosure more challenging in multiparty settings (Cooney et al., 2020).

Addressing these complexities and building computational models capable of handling multiparty conversations has been an important agenda within Natural Language Processing (NLP). Researchers have made significant efforts in this area by adopting various approaches, broadly categorized into conversational context modeling and conversational component modeling (Gu et al., 2022b). Conversational context modeling focuses on models' overall understanding capability of multiparty conversation. Key methods include dialogue discourse parsing, which defines relationships between utterances by analyzing discourse structures (Afantenos et al., 2015; Wang et al., 2021); information flow modeling, which captures conversational dynamics using graph-based encodings (Hu et al., 2019; Lee and Choi, 2021); and self-supervised training, which learns representations from unlabeled data (Gu et al., 2021; Li and Zhao, 2021). Conversational component modeling, on the other hand, decomposes multiparty conversations into core tasks related to WHO speaks, WHAT is said, and WHOM is addressed (Gu et al., 2022b). For this category, approaches include: next-speaker prediction, which relies on turn-taking patterns (de Bayser et al., 2019) or a combination of turn-taking and utterance semantics (Gu et al., 2021); speech content generation, approached through retrieval-based (Ouchi and Tsuboi, 2016) or generation-based methods (Gu et al., 2022a); and addressee recognition, which can be approached via methods such as dialogue disentanglement (Elsner and Charniak, 2008) or explicit addressee modeling (Ouchi and Tsuboi, 2016; Gu et al., 2021). All these efforts lay the groundwork for designing CAs capable of understanding multiparty conversations.

Beyond purely linguistic signals, incorporating multimodal inputs is another typical approach to help CAs participate in multiparty conversation. Early systems, such as those developed by Bohus and Horvitz (Bohus and Horvitz, 2011a, 2009a,b, 2011b), along with recent AI-enabled social robots (Addlesee et al., 2024), demonstrate the benefits of leveraging non-verbal cues like gaze and F-formation (Xu et al., 2013; Marshall et al., 2011). These cues help models better predict user attention and engagement, enabling more effective turn-taking and smoother conversational integration. However, such multimodal signals are primarily available to embodied agents (e.g., social robots or virtual avatars) and remain inaccessible to the vast majority of text- or voice-based CAs. This paper focuses primarily on text-based multiparty interaction, believing that understanding and designing for this prevalent modality is a crucial and necessary starting point for systematically advancing CA design.

More recently, the advent of Large Language Models (LLMs) has offered a new paradigm for text-based CAs. Pre-trained on massive and diverse text data, LLMs like GPT-4 (OpenAI, 2024) and DeepSeek-V3 (DeepSeek-AI, 2025) demonstrate remarkable abilities in understanding context, generating coherent text, and capturing subtle linguistic nuances over long sequences. These strengths suggest LLMs are uniquely well-suited to tackle the intricacies of multiparty conversations. Indeed, Tan et al.'s evaluation of GPT on multiparty conversation datasets, has confirmed this potential, showing capabilities particularly in tasks including addressee prediction, response generation, and response selection (Tan et al., 2023). For instance, Liu et al. proposed the *Inner Thoughts* framework (Liu et al., 2025), equipping LLM-based CAs with a simulated internal thinking process and intrinsic motivation to trigger proactive, human-like participation in multiparty conversations. Similarly, Houde et al.'s *Koala* method leverages an LLM to generate candidate responses and evaluate their conversational value, enabling the CA to decide when to participate in group discussions (Houde et al., 2025). Therefore, designing CAs for multiparty conversation is increasingly becoming less about sheer technical feasibility, but more fundamentally, a design problem. This involves identifying what mechanisms are missing in current CAs compared to human-human multiparty conversation, and determining how an agent should appropriately participate in a dynamic group of humans.

2.3. Conversational agents in multiparty contexts

Driven by the growing interest in integrating AI into the pervasive group interactions common in social and professional settings, a dedicated line of research has emerged focusing on developing and evaluating CAs specifically intended for or deployed within multiparty environments. These agents are also referred to as polyadic CAs (Kuhail et al., 2025; Zheng et al., 2022).

Existing polyadic CAs deployed in multiparty settings are predominantly lightweight, task-oriented chatbots that facilitate collaboration. Typical functions include coordinating schedules (Cranshaw et al., 2017), managing workflows (Toxtli et al., 2018), and providing on-demand information (Avula et al., 2018). These bots are now routinely embedded in platforms such as Slack, Discord, and Twitch to deliver such services (Seering et al., 2018, 2019). Yet most systems either respond passively to explicit one-to-one requests or broadcast generic updates to the entire group (Bucher et al., 2024), making information exchange essentially one-way. Consequently, it remains difficult to recognize their capacity for real-time engagement within dynamic multiparty conversations.

Later work has expanded the scope of polyadic agents into more dynamic collaboration, particularly as facilitators and moderators for discussion (Kim et al., 2020, 2021) and emotional support (Benke et al., 2020; Peng et al., 2019). For example, Kim et al. developed a chatbot that proactively manages discussion timelines, encourages balanced participation, and synthesizes group opinions (Kim et al., 2020). While this type of CA has been shown to improve group engagement and

collaborative task outcomes, the chatbot itself acts more as an "external coordinator" or "overseer." It directs the conversation rather than participating in it, thus lacking genuine engagement. The combined limitations of this line of research — namely, the lack of engaging capability in many current systems and a confined design focus on merely functional roles — reveal a significant, yet uncharted, design space for polyadic CAs. Specifically, many current CAs were designed heavily on resolving human-human interaction challenges (e.g., inefficient communication or uneven participation (Kim et al., 2020, 2021)), while overlooking the agents' social capabilities. This reflects a broader conceptual divide: current designs treat CAs as tools *for* groups rather than social entities *within* groups, despite growing recognition of their potential as relational partners (Seering et al., 2019).

While this remains a persistent challenge, recent research has begun to push these boundaries. For instance, Liu et al.'s *Inner Thoughts* framework (Liu et al., 2025) and Houde et al.'s *Koala* system (Houde et al., 2025) both exemplify efforts to design CAs as "another member" in conversation, rather than as mere tools or external facilitators, emphasizing proactive and more autonomous engagement. Despite these advancements, the full scope of designing CAs to genuinely participate as integrated members in complex human multiparty interactions remains largely unexplored. In particular, questions such as how CAs can become socially integrated into groups and how they should behave in fluid multiparty settings are often neglected. The Computer as Social Actor (CASA) (Nass et al., 1994) paradigm reminds us that people instinctively apply social rules and heuristics to computers, treating them as social agents even if they are not explicitly designed to be. Thus, ignoring these questions risks both sub-optimal designs and misleading evaluations. However, most existing discussions, for example, about social characteristics (Chaves and Gerosa, 2021) or persona of CAs (Sutcliffe, 2023; Sun et al., 2024), remain largely restricted to dyadic contexts. Even Houde et al.'s study (Houde et al., 2025) examined user preferences for CAs' proactivity in their group ideation settings, finding that a reactive mode was preferred. Yet initiative is only one facet of the broader question of socially acceptable CAs in multiparty settings. We believe a much broader design space remains to be charted, including the social roles, interactional behaviors, and agent personas users truly envision and expect. This gap motivates our semi-structured interviews (Study 2) to investigate user expectations of CAs in more general multiparty conversations.

3. Research questions

In summary, we still lack a clear picture of the complexities that arise in multiparty conversations and how participants successfully manage them. This limited understanding severely constrains our design approach, leading to multiparty conversational agents that function primarily as simple tools or external coordinators of group dynamics. To bridge these gaps and advance the design of CAs that engage as genuine participants in multiparty conversations, we therefore investigate the following research questions:

- **RQ1.** How do people participate and contribute to multiparty conversations, and what underlying mechanisms shape these interaction dynamics?
- **RQ2.** What capabilities and design requirements are essential for CAs to effectively navigate multiparty conversations, given the core mechanisms identified in RQ1?
- **RQ3.** What social roles, interactional behaviors, and agent personas do users envision and expect for CAs in multiparty contexts?

4. Study 1: retrospective think-aloud on multiparty conversations

To address **RQ1** by gaining deeper insights into dynamic multiparty conversations, we revisit human-human interaction, which is rich with intricate mechanisms governing natural communication. These insights are crucial for understanding not only how multiparty conversations unfold, but also the underlying reasons and mechanisms that govern their dynamics, thereby illuminating what prevents current CAs from engaging as genuine, seamless participants in group interaction (in service of **RQ2**). Instead of relying solely on descriptive analyses of micro-level interactions within conversation histories, we prioritize participants' internal reasoning behind their conversational actions, using a retrospective think-aloud approach (Charters, 2003). This method bridges the gap between observable behaviors and tacit cognitive and social reasoning, enabling us to uncover the essential mechanisms that shape natural multiparty interactions.

4.1. Study environment

We simulated real-time, non-face-to-face, text-based multiparty conversations using the online collaborative platform Slack. While multiparty conversations occur across various modalities, we specifically focused on this text-based format for our initial investigation. This modality is particularly ecologically relevant because it is ubiquitous in daily life and presents unique challenges for CA design. Specifically, text-based interactions often assume or empirically exhibit sustained, long-term engagement, unlike the typically shorter, turn-based exchanges common with voice-based CAs (Porcheron et al., 2018; Bickmore and Picard, 2005). Furthermore, the inherent lack of non-verbal cues in text-based communication makes nuanced social participation more challenging for CAs. Therefore, insights derived from this setting are crucial for informing the design of future agents capable of engaging as social entities in similar text-based environments. Focusing on this modality provides a critical and manageable starting point for research before tackling the complexities of richer, multimodal settings.

We used Slack simply as a stable, widely familiar conduit for real-time group chat. To ensure a focused text-based experience and avoid specific Slack features that might introduce feature-specific complexities, we explicitly reminded participants in the briefing not to use message threading and forwarding. Participants were permitted to use common text-based chat features such as mentions and emoji reactions.

4.2. Study design and procedure

Participants were recruited in groups of three and assigned to a shared Slack channel. To capture a broader spectrum of interaction dynamics and participant behaviors, each group engaged in two types of conversations — task-oriented and casual social talk — since they were regarded as serving different purposes and engaging distinct dynamics (Clark et al., 2019). Each type comprised two topics, resulting in four total conversation topics. Here we labeled task-oriented topics as T1 & T2 and casual social topics as C1 & C2:

- **T1**: *Plan a trip to Japan*
- **T2**: Debate on *"Does technology make life better?"*
- **C1**: Chat on *"What did you do last weekend?"*
- **C2**: Chat on *"What would you do if you had a million dollars?"*

Every group participated in four chat sessions, each followed by a retrospective think-aloud. Each session covered a different topic. As shown in Fig. 1, in each session, participants first had a conversation on the assigned topic. Immediately afterwards, the complete message log was screen shared, and participants were asked to walk through it turn by turn. For every utterance they had contributed, they verbalized the thoughts, considerations, reasoning processes, or decision strategies that motivated that action. This transcript-guided retrospective think-aloud kept recall demands low and exposed the cognitive processes underpinning the just-completed interaction. Each conversation was limited to 5 minutes to ensure the accuracy of memory in retrospect and reduce the workload of the think-aloud process. To ensure the conversations remained natural despite the time limit, we instructed participants to disregard the time constraint and focus on interacting as they normally would until we stopped the conversation.

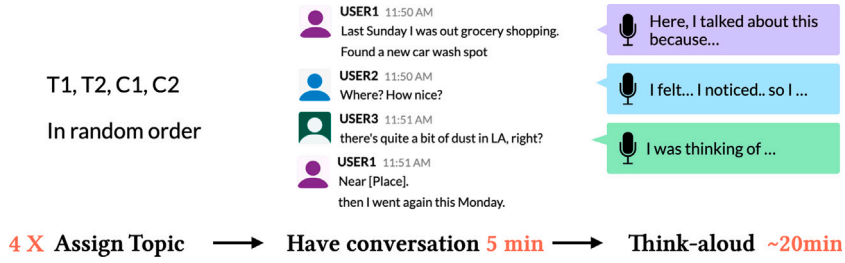


Fig. 1. Procedure of the retrospective think-aloud study.

Table 1
Group demographics of think aloud participants.

Group*	1	2	3	4	5	6	7	8	9	10
Gender†	2F-1M	1F-2M	3M	2F-1M	2F-1M	1F-2M	2F-1M	1F-2M	3F	3M
Familiarity‡	3	3	3	2–1	3	2–1	3	1–1–1	2–1	1–1–1

* In the following sections, participants will be denoted using a specific format. For example, G2P3 refers to Participant 3 from Group 2.
† For Gender: F = Female, M = Male.
‡ For Familiarity: 3 = all participants knew each other prior to the study; 2–1 = two participants knew each other while one did not; 1–1–1 = none of the participants knew each other.

The order of the four topics was counterbalanced across groups to mitigate potential fatigue, order, and carryover effects (Kirk, 2013), ensuring that the sequence of tasks did not systematically bias the results. Overall, each group's entire study lasted about 1.5 hours.

4.3. Participants

In this study, a total of 10 groups, comprising 30 participants, were recruited through social networking platforms. Of these participants, 14 were based in the United States, 13 in China, 2 in Japan, and 1 in Canada. They were 21–36 years old (M = 26.5, SD = 3.3) and represented diverse professions (students, engineers, designers, and finance professionals), suggesting regular exposure to multiparty conversations. To minimize tool-use friction, we confirmed that each participant had used Slack before the user study so they could operate the interface without additional training.

As summarized in Table 1, each group consisted of three participants and differed in both gender composition and prior familiarity: 4 groups consisted of 2 females and 1 male (2F1M), 3 groups consisted of 1 female and 2 males (1F2M), 2 groups were all male (3M), and 1 group was all female (3F).

Regarding participant familiarity, 5 groups were composed of individuals who all knew each other before the user study, 2 groups included members who were all unfamiliar with one another, and 3 groups had two participants who knew each other plus one unfamiliar participant. Having different gender ratios and familiarity levels ensures that our study is not confined to a narrow set of social dynamics, thus supporting broader generalizability of the findings.

Each participant was compensated around 14 USD in local currency.

4.4. Data collection and analysis

With approval from our Institutional Review Board (IRB), we downloaded the conversation logs and audio-recorded retrospective think-aloud sessions. Audio recordings were transcribed verbatim, resulting in approximately 16 hours of transcribed data.

We conducted inductive thematic analysis to systematically identify patterns in participants' reasoning about multiparty interaction strategies. The analysis followed Braun and Clarke's six-phase framework (Braun and Clarke, 2024): Two researchers first independently reviewed transcripts and conversation logs to gain immersion in the dataset. Then, researchers generated codes inductively while highlighting excerpts relevant to RQ1. Afterwards, codes were grouped into preliminary themes through iterative discussions. In the review and refinement phase, a collaborative session was held to resolve coding discrepancies (e.g., merging similar codes, distinguishing conflicting codes) and refine themes against the full dataset. Later, themes were articulated with clear definitions and mapped to research questions. Finally, representative quotes were selected to illustrate each theme, ensuring coverage of both task-oriented and casual social contexts.

5. Findings of study 1 - key mechanisms for engaging multiparty conversation

Drawing on retrospective think-aloud data from our study of text-based multiparty chats, we observed a notable convergence in how participants reasoned about their conversational actions. By triangulating participant reasoning, conversational logs, and prior literature on group dynamics, we exploratorily synthesize 5 key mechanisms: turn-taking, thread, relevance, presence, and role management. Presented as a cohesive set, we propose that these mechanisms collectively explain how individuals organize conversational flow and ensure interactions that are contextually appropriate, meaningful, and engaging. Further, they offer human-centered inspiration for designing CAs to engage as more genuine participants in multiparty conversation. Each mechanism is elaborated in the subsections that follow.

5.1. Turn-taking management

Echoing the literature on turn-taking theory (Sacks et al., 1978), two primary modalities of turn-taking — **turn-allocation** and **self-selection** — were consistently identified in their conversation logs and think-aloud content.

To quantify the prevalence of these turn-taking modalities, we first analyzed the turns across all 40 conversation sessions in our study, identifying each as either *self-selected* or *allocated*. As illustrated in Fig. 2, a substantial majority of turns were taken voluntarily through self-selection (M = .86, SD = .06). Specifically, casual chats (C1 & C2) exhibited a significantly higher proportion of self-selected turns ($C1$: M = .90, SD = .06; $C2$: M = .91, SD = .05) compared to task-oriented

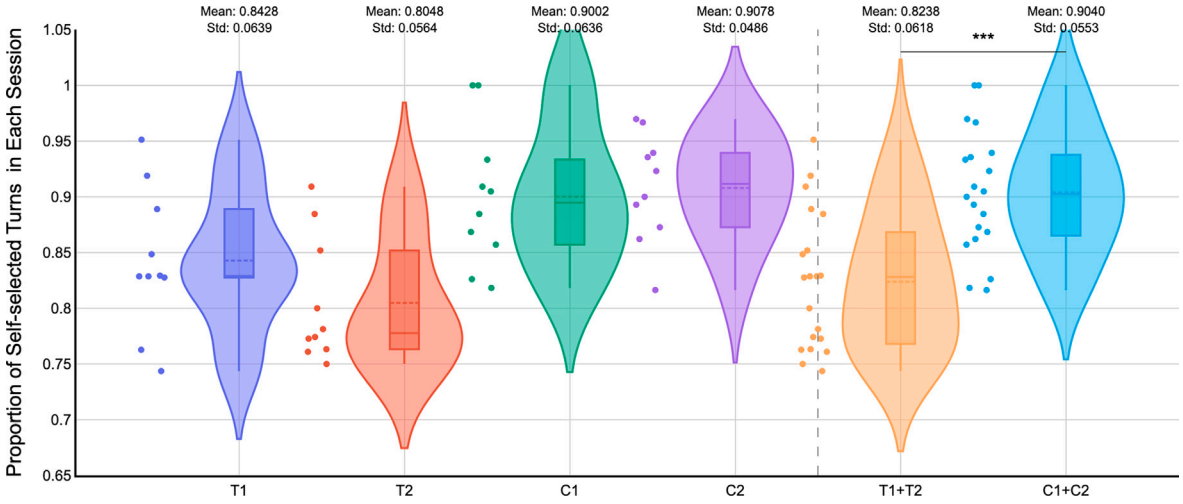


Fig. 2. Proportion of self-selected turns across conversation topics.

chats (T1 & T2) ($T1$: $M = .84$, $SD = .06$; $T2$: $M = .80$, $SD = .06$). When collapsing the data by conversation type (task-oriented vs. casual), an independent samples t-test indicated this difference was statistically significant ($t(38) = -4.4$, $p < 0.001$). These quantitative patterns echo participants' qualitative reasoning.

5.1.1. Turn-allocation in multiparty conversation

Turn-allocation refers to the process by which conversational turns are assigned to participants. In dyadic conversations, this process is relatively simple: when one person finishes speaking, the other is naturally assigned the next turn. This transition is often implicit and occurs with minimal signaling. In text-based multiparty conversations, however, turn allocation frequently demands explicit signals such as @mentions, direct questions, or naming specific individuals. This explicit approach can simplify coordination in multiparty contexts — particularly when specific expertise or clarification is needed, or when the conversation risks fragmentation.

> "I'm mentioning him (P2) here because I think he knows more about this and I'm more curious about his opinion" *[G1P1]*

Turn-allocation is the straightforward reason why participants join the conversation. Participants frequently described feeling obligated to respond when directly addressed, even when unprepared:

> "At that point, I was cued, so I had to say something, even though my mind was completely blank."*[G4P1]*

Participants also noted instances of "turn-allocation escape", where designated recipients intentionally or unintentionally declined their allocated turn:

> "I wasn't trying to ignore her on purpose. I just found it really hard to answer at that moment. Later, when they switched topics, I just went along with it."*[G5P2]*

This dynamic sometimes leads to a brief pause or confusion, highlighting how turn-allocation is not foolproof. It still relies on individuals being willing to accept or reject the allocated turn, which leads to the most dominant way of turn-taking in the multiparty conversation — self-selection.

5.1.2. Self-selection in multiparty conversation

Self-selection occurs when a participant voluntarily claims a turn without being explicitly invited. It requires participants to monitor multiple potential entry points and participate in the conversation based on their willingness.

In dyadic conversations, self-selection is less frequent because it often leads to interruptions or overlaps, which may require further conversational repair (Schegloff et al., 1977).

In our study, as shown in Fig. 2, over 86 % of turns were taken through self-selection, rising to over 90 % in casual chats. Participants emphasized that the text's asynchronous nature facilitates this dominance.

> "I usually just speak my mind directly, and typing it out together doesn't really cause any problems anyway."*[G5P3]*

Participants frequently employed output buffering when self-selecting. They internally took the turn first but were more mindful about the coordination in the conversation.

> "(At that moment) I had already typed my response, but when I saw that P3 was also typing, I decided to wait and see what he wanted to say."[G3P2]

Some participants noted that temporal gaps and topic boundaries prompted them to take the turn. For example, when they observed a pause in the conversation or recognized that a topic had concluded without anyone else speaking, they perceived a social vacancy and seized the opportunity to ensure the conversation progressed smoothly.

Additionally, participants highlighted expertise considerations (e.g., *"I know a lot about this topic and should share my knowledge"*) and time-sensitive corrections (e.g., *"There is a factual mistake that could lead the conversation in the wrong direction"*) as factors that created a sense of urgency to take the turn.

A few participants also mentioned the reasoning process nuances of task-oriented conversation and casual social talks.

> "In task-oriented situations, I tend to focus more on 'what we need', but in everyday conversations, I pay more attention to 'what I want to express'."*[G8P1]*

Finally, participants emphasized the importance of establishing an environment conducive to self-selection. They intentionally fostered an atmosphere where all members felt comfortable to self-select turns, thereby enhancing the conversation's overall effectiveness and harmony. For example, participants employed proactive turn-yielding strategies, such as concluding their statements with questions like *"What do you guys think?"*, or utilized non-competitive responses, including adding reactions and providing short affirmations through backchanneling. These act as visual affordances for others to self-select.

5.2. Thread management

In contrast to dyadic conversations, which typically focus on a single topic at a time, multiparty conversations often involve multiple ongoing topics or sub-conversations simultaneously. This parallelism adds complexity but also enriches the interaction. While this is a well-documented characteristic of multiparty conversation (Aoki et al., 2006) and a key consideration in computational modeling of discourse structure (Mahajan, 2023), our study sought to delve deeper into how participants, from their own perspectives, actively navigate and manage these threads. In our data, we followed the full life cycle of conversation threads — their initiation, progression, and closure — and specifically captured the underlying reasoning that guides participants' decisions at each stage.

5.2.1. Initiation

Thread initiation occurs dynamically in text-based multiparty conversations. We identified through our thematic coding three primary, interconnected reasons guiding participants' decisions to initiate new threads: cognitive divergence, task decomposition, and social scaffolding. These emergent strategies enable groups to balance conversational breadth with depth, and to foster inclusivity while maintaining overall conversational coherence.

For cognitive-driven initiation, participants often introduce new topics or subtopics through associative thinking or external stimuli. For example, with associative thinking, a message about travel logistics might trigger a memory of local cuisine, prompting a parallel thread about food. External stimuli, such as a new message or notification participants receive from a smartphone during the study, can also prompt new topics. These shifts often reflect attentional capture, as participants branch discussions organically:

> "I brought up this topic because I just saw a news pop-up and thought other two might find it interesting." *[G8P2]*

Such diversions are especially common in casual contexts, where free-flowing digressions enhance engagement.

A second driver, task-driven initiation, arises from the need to decompose complex objectives into manageable sub-threads. In task-oriented settings, participants frequently divide discussions into parallel strands to optimize efficiency. For instance, when planning a trip, they might simultaneously discuss dates, budgeting, and itineraries by distributing tasks among themselves. This initiation was reported to expand the conversational width and reduce cognitive load in task-processing.

Furthermore, participants described intentionally deepening conversations by initiating question-and-clarification chains, where their questions would deliberately spawn dedicated sub-threads to resolve ambiguities or fill information gaps critical for task progression.

> "I asked this question to understand exactly what P1 was referring to; otherwise, I couldn't help deal with [...]" *[G4P3]*

This process of seeking clarification not only involves the original questioner and respondent but, by its nature, often encourages others to join and contribute, thereby collaboratively building a shared and more detailed context.

Finally, social scaffolding motivates thread creation to balance participation and mitigate conflict. Participants create dedicated threads to engage quieter members or steer conversations away from sensitive topics. For example, one participant described creating a side thread to avoid discomfort:

> "Talking about these (drugs and gambling) makes me uncomfortable, so I wanted to use that to steer the conversation in a different direction." *[G10P2]*

By strategically introducing new threads for these social reasons, participants aimed to maintain a positive group dynamic and ensure that all members felt included and at ease within the conversational environment.

5.2.2. Progression

Participants in multiparty text-based conversations navigate and progress through threads using several strategies that enable them to manage multiple concurrent discussions effectively. This progression involves choosing and affiliating with threads, prioritizing threads based on perceived importance, and dynamically bridging across threads.

In text-based platforms like Slack, participants have the flexibility to engage with multiple threads simultaneously. Also, the ability to scroll back and review previous messages reduces their cognitive load when affiliating with different conversation threads. Although participants complained that this flexibility comes at the cost of a less ordered conversational sequence and makes reviewing in think-aloud mentally demanding, it did enhance participation by allowing individuals to contribute where they feel most valuable while staying informed about other ongoing discussions.

> "I was responding to P1 here, last sentence was for P2. Because they are interested in different issues, so I ended up chatting with them simultaneously. I guess it is pretty common (in online chatting)." *[G8P3]*

Participants reported that they selectively chose which threads to engage with based on personal interest, relevance, or expertise, similar to how they decide on turn-taking with self-selection.

They often perceive certain threads as more dominant or higher in priority, leading to a hierarchy of threads. This influences how they allocate their attention and contributions. Factors that affect thread prioritization include the urgency of the topic, collective interest, the group's objectives, and personal responsibility. Different participants may prioritize different threads, but these priorities tend to converge more in task-oriented conversation. For example, participants tacitly set aside chit-chat and then debated on AI-writing tools in topic 2.

> "I feel it's more important to clarify my position first... So I stop joking around"*[G8P2]*

The affiliation with multiple threads also emerges as cross-thread bridging, where participants link ideas and information from one thread to another. This practice reflects that participants maintain a continuous awareness of inter-dependencies, enabling groups to maintain consistency across fragmented discussions.

5.2.3. Closure

When no participants affiliate with a thread, it is considered closed. However, participants explained that this closure is not necessarily permanent; they might consciously decide to reactivate a thread if new relevant information arises or if they feel a strong personal motivation to revisit a topic.

> "I don't know if she'll get annoyed, haha, but I suddenly became really curious about what was just said regarding her parents, so I brought it up again." *[G5P3]*

More commonly, participants described an implicit, often unstated, agreement within the group to limit the number of active threads. This self-regulation, aimed at maintaining conversational coherence and preventing cognitive overload, frequently involved individuals deciding to disengage from less active or less pressing threads, or to hold back contributions that might further fragment the discussion. Without explicit coordination, they reported recognizing when it was appropriate to allow a thread to conclude.

> "I wanted to keep sharing something, but it felt too chaotic to talk about right now, so I didn't rush to bring it up." *[G2P3]*

Finally, participants also noted that thread lifecycle patterns differ between task-oriented and casual social contexts.

> "I feel that in task-oriented conversations, once we've discussed something, that's the end of it. But in casual talks, things are more

random, we might suddenly stop chatting this and then pick it up again at a random point." *[G1P1]*

This contextual adaptability in deciding when and how threads conclude underscores that there is a dynamic balance between structural flexibility and collaborative intentionality in thread management.

5.3. Relevance management

In our study, the concept of **relevance** was frequently mentioned by participants. Relevance operates as the primary cognitive filter through which participants navigate multiparty conversations, guiding decisions to engage in threads or self-select turns.

Relevance, in this context, refers to how well a participant's contributions, whether through utterances or reactions, align with the ongoing topics or goals of the conversation. As a widely discussed concept in previous literature, relevance is considered a basic normative ideal of conversation. Like in Grice's Cooperative Principle for successful communication, *"Be relevant"* is one of the four maxims (Grice, 1975). Sperber and Wilson defined relevance as *"newly presented information being processed in the context of information that has itself been previously processed"* (Wilson, 2017) and proposed that communication is driven by a search for the most relevant information with the least cognitive effort. This theoretical grounding was echoed in participants' own reasoning:

> "If he's talking about something I've encountered or experienced before, I'd want to join in and discuss it." *[G6P3]*

The search for relevance operates through a cognitive interplay between working memory (tracking real-time discourse) and long-term memory (activating prior knowledge/experiences). Participants instinctively matched in-situ conversational cues to their personal knowledge bases, interests, and social goals, triggering engagement desires.

> "That actually reminded me of the car wash P2 and I went last weekend, but I was worried P3 might think it's irrelevant and get lost." *[G6P1]*

However, relevance judgments are inherently subjective, necessitating theory of mind (ToM) (Frith and Frith, 2005) to coordinate contributions. Participants continuously assessed whether internally relevant content would align with others' mental states. As illustrated in the quote above, people with similar experiences are more likely to understand and identify with relevant content in a conversation. In contrast, those who lack similar experiences may have difficulty relating certain information to their reality and consequently find the topics lacking in relevance. In multiparty conversations, participants tend to seek shared relevance:

> "She said something that really resonated with me, so I felt it was important to express that." *[G1P3]*

When a participant adds something relevant, it does not sit in isolation — it reshapes the immediate context for other interlocutors involved. As subsequent participants seek to make their own contributions relevant, they naturally find common ground with the previous speaker. This ongoing cycle of relevance-seeking gradually constructs shared understanding and fosters mutual understanding among participants, and creates a contextual momentum. This process also aligns with findings from prior work where establishing such common ground was identified as an integral feature of good conversation (Clark et al., 2019).

Of course, strict relevance in conversation is not necessary in all cases. We also observed instances where participants made utterances that were not directly relevant to the immediate topic but were relevant to the participants themselves in a broader context beyond the conversation. We refer to this as *loosened relevance*. This type of contribution is more likely to occur in social conversations, where participants sometimes diverge from the main topic for humorous or playful purposes. Several participants noted that introducing off-topic remarks can lighten the mood or create a sense of relaxation, helping to strengthen social bonds within the group.

> "My thoughts jump around quite a lot, but luckily, everyone here knows me well and accepts it. In other situations, I'd probably think twice before speaking. I actually got criticized for this in my work. " *[G4P3]*

The success of loosened relevance often depends on the social structure of the conversation — participants with closer relationships or higher social status within the group may be more comfortable introducing off-topic remarks without negative repercussions.

This loosened relevance carries certain risks of not being tolerated. While it can foster a more relaxed and socially cohesive environment, it can also derail the conversation or cause confusion if the shift is not well-received or if the group is focused on a more goal-oriented discussion. Nevertheless, loosened relevance does not abandon the principle of relevance; rather, it reorients it. Participants are still engaged in the process of building shared relevance within the group. In this way, relevance persists as the driving force of interaction, even when the conversation occasionally takes unexpected turns.

5.4. Presence management

Social presence, the perceived sense of "being there" with others in mediated communication, is a well-established concept in group interaction research and has been extensively investigated (Oh et al., 2018). Our data highlight a related but more dynamic phenomenon we term **presence management**. Rather than merely being present, users actively tune how much they and others should be seen at any moment. Presence management, therefore, is about how individuals strategically signal their own and others' engagement or disengagement to control visibility and participation within the conversation. This active mechanism was observed to manifest in two key forms: participants' management of their own presence (**self-presence**) and their efforts to influence the presence of others (**peer-presence**).

5.4.1. Self-presence

In text-based conversation, participants reported primarily two strategies for enhancing their self-presence. More frequent turn-claiming was often described as the most direct or immediate approach, and alongside this, backchanneling was commonly cited as another dedicated method for signaling visibility and engagement.

We observed a significant amount of backchanneling in these conversations, which included sending short affirmations (e.g., *"yeah", "I agree"*), using short interjections (e.g., "?!", "wow", "lol"), or sending and reacting with emojis. We quantified backchanneling across the 40 sessions of conversation we logged, identifying 217 back-channel turns out of 1378 turns overall. Per session, the proportion of back-channels ranged from 0 to 0.34, clustering around 0.15 of all turns. Fig. 3 shows the distribution for each group–topic combination.

While present, this quantitative data suggests a relatively low level of overt backchanneling in these text-based multiparty conversations. Our qualitative insights help explain this pattern: rather than a lack of engagement, the lower frequency appears to stem from a perceived friction in the conversation medium. Participants explicitly complained about the limited effective ways to signal presence without disrupting the flow, leading to a gap between their intent to affirm and their actual behavior. Indeed, many reported backchanneling less than they intended.

> "I want to show I was actively listening, so I reacted a bunch of emojis. In my daily life, I would speak more (affirmations), but typing those responses feels like adding noise to the conversation." *[G6P2]*

Additionally, in lively multiparty conversations with different threads happening simultaneously and at a fast pace, it can be difficult for participants to clearly indicate whom or which thread their backchannel is directed toward. This targeting challenge, they reported, also leads to ambiguity in interpreting backchanneling intentions —

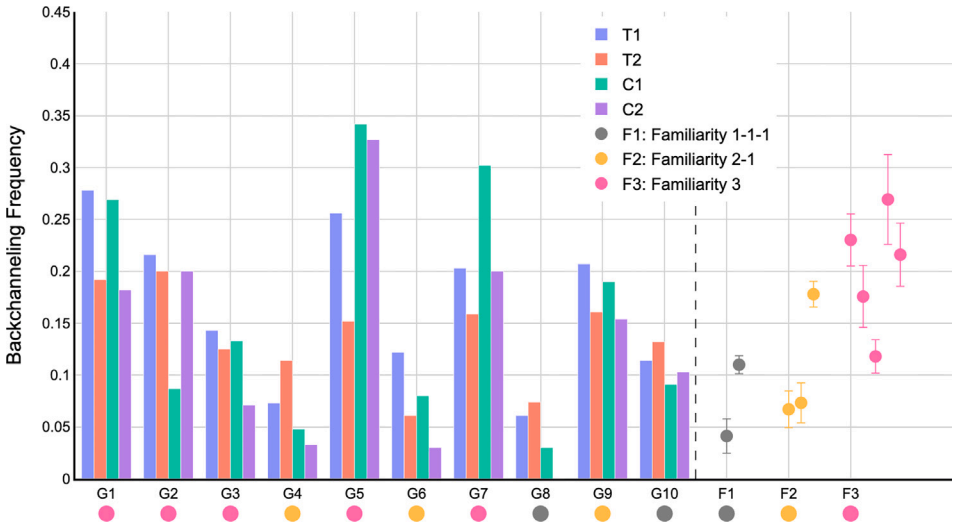


Fig. 3. Backchanneling frequency distribution by group and familiarity level.

making it unclear whether the sender intends to signal active presence or prefers to show less of it.

> "I'm actually pretty worried that others might misinterpret my backchanneling (here). I want to show that I'm being proactive, but to others, it might look like I'm being perfunctory." *[G9P2]*

This perceived risk of misinterpretation, as highlighted by G9P2, might be particularly heightened in groups with lower or mixed familiarity. Our descriptive analysis of backchannel frequency, illustrated in dot plot of Fig. 3 offers some tentative support for this: groups with lower familiarity appeared to exhibit less frequent backchanneling. Although the small number of truly independent groups limits the statistical power for a robust significance test, our aggregation of the data (averaging backchannel frequency per group across its four topics and plotting by familiarity category) may suggest this potential pattern.

Further contributing to the potential for misinterpretation is the versatile nature of backchanneling itself. Beyond signaling active engagement, participants also indicated that backchanneling can be employed as a strategy to maintain only a minimal presence. This might occur when individuals decide to reduce their overt participation but still wish to offer slight reactions without fully committing to a turn or deeper engagement.

While such reduced activity could be perceived by others as disengagement, some participants indicated that this conscious choice of passive self-presence management serves strategic and positive functions, such as simplifying a complex discussion or ensuring others have adequate space to contribute. As G1P3 explained their reasoning for stepping back:

> "Again, I feel like I've talking too much and keep outputting my stuff. I need to give them a chance to share their things." *[G1P3]*

This thoughtful calibration of one's own presence to create space for others, demonstrates that self-presence management is often deeply intertwined with an awareness of the group's overall conversational health. In addition to regulating themselves, participants also described tactics for shaping the visibility of their partners, a theme we turn to next.

5.4.2. Peer-presence

Participants also discussed their conscious strategies for managing peer-presence, focusing on how they perceived and subsequently attempted to alter others' engagement levels. For instance, they might actively allocate turns to quieter individuals, asking questions like, *"What do you think about this?"* to bring them into the conversation. This strategy was particularly noted as a way to include individuals in the conversation in mixed familiarity groups, as G9P1 explained their rationale:

> "P3 is new to us, so I tagged her, didn't want her to feel alienated." *[G9P1]*

Beyond such explicit inclusion, employing "implicit inclining" was another subtle strategy participants reported. This involved consciously tailoring topics or directing replies in ways they believed would resonate with a quieter individual's interests, thereby aiming to lower social barriers and encourage their participation. G3P3 described this approach:

> "I feel like P2's being too quiet about this topic, so I bring up more things I think he'll enjoy, hoping he'll participate more." *[G3P3]*

Interestingly, behaviors aimed at negatively managing peer-presence (e.g., intentionally trying to lower others' presence) were not reported. Participants indicated a stronger tendency to focus on enhancing their own presence constructively or fostering an inclusive environment.

The management of peer-presence and self-presence appears to be a balanced act. Rooted in social norms and emerging quite naturally from the participants' accounts, this dynamic interplay mechanism plays an important role in fostering inclusivity and promoting more balanced contributions within multiparty conversations.

5.5. Role management

Roles in conversation refer to the set of behaviors, expectations, and responsibilities that individuals adopt or are assigned during an interaction (Goffman, 2023). These roles range from basic ones, such as speaker and listener, to more functional roles like moderator or facilitator, and context-dependent social roles. Unlike dyadic conversations, where roles are relatively stable and predictable, roles in multiparty conversations are fluid and often negotiated in real-time — a dynamic process participants described as both strategic and instinctive, as G1P1 articulated when explaining their decision to adopt an initiative-taking role:

> "I know that none of us are typically the type to proactively steer the conversation, so I decided to take the first step and initiate the topic." *[G1P1]*

Participants typically enter conversations with an awareness of their own roles and the roles of others, informed by factors such as status, expertise, relationships, or social hierarchies within the group. This initial

role recognition is shaped by a combination of self-concept (an individual's beliefs about themselves) and social comparison (evaluating oneself relative to others).

When participants are unfamiliar with one another, this pre-existing role identification process becomes difficult. However, roles stabilize over a few conversational turns, inferred from behaviors and contributions during the onboarding phase of the conversation.

> "At first, no one took charge. But after P3 started summarizing key points, we all treated him as the de facto moderator." *[G10P2]*

As the conversation unfolds, roles are not static but evolve and shift according to the flow of the conversation and the group's needs. Role emergence occurs naturally as new conversational threads are introduced or as the conversation transitions into a new stage.

> "(topic switch to mortgage) I don't really have much knowledge about this, so I just went along with it and mostly listened to everyone else talk." *[G2P2]*

Participants often shift their roles based on the changing focus of the conversation, taking on roles that better align with their expertise or the situation at hand. The capacity to navigate these role transitions fluidly is a key marker of successful multiparty conversations.

Roles are not singular, and participants may embody different roles in different threads or even carry multiple roles within the same thread. The former situation, for example, a participant as a geography enthusiast might serve as an information provider when discussing travel destinations but as an information seeker in conversations about budgeting. The latter situation, such as one participant carries both task-oriented and socio-emotional roles in a work meeting, but struggling to balance them:

> "I wanted to keep the discussion on track, but I also didn't want to be too bossy." *[G1P1]*

The struggle expressed by G1P1 highlights one type of role conflict — an internal tension when balancing differing role demands. Beyond such self-conflict, issues in role management also arose from the mismatch between self-perception and others' expectations.

This is often because, in most cases, conversational roles are implicit. Only in a few task-oriented conversations that require division of labor are roles explicitly declared or assigned. This can result in hesitancy to contribute due to uncertainty about one's role.

Role adaptation is an essential strategy for resolving these conflicts and responding effectively to the dynamics of multiparty conversations. However, recognizing when and how to adapt roles during a conversation is not trivial. It requires cognitive flexibility—the ability to shift perspectives and respond to new information. Some participants reported that this flexibility can be constrained in social environments characterized by strong cultural norms or rigid hierarchies.

> "I feel comfortable being this confrontational because we're all pretty equal. If I were talking to my boss, I'd probably be more compliant." *[G5P3]*

Effective role adaptation also requires responsiveness, demonstrated through behaviors like active listening and empathy. Participants engage in processes akin to theory of mind, enabling them to understand, predict, and share the feelings of others. The capacity to adapt roles fluidly allows participants to meet the evolving needs of the conversation and the group. By adjusting their roles, they can contribute more effectively, foster better collaboration, and enhance the overall quality of the interaction.

6. Study 2: semi-structured interview

While Study 1 identified the key interactional mechanisms of human conversation, simply replicating these functions technically does not guarantee that CAs will be accepted into the human social fabric. Indeed, prior work establishes that user relationships and interactions with technology are driven heavily by people's pre-existing beliefs and expectations (Nass et al., 1994; Grimes et al., 2021). Therefore, to complement insights from our first study, we conducted Study 2 to understand what constitutes a believable and socially integrated CA in multiparty contexts from a user-centric perspective. This study employed semi-structured interviews to directly investigate our RQ3, focusing on the specific expectations and desired persona traits.

6.1. Study design and interview procedure

Interviews began with participants being asked to recall a recent or memorable multiparty conversation they had been part of. They described its basic details, including the group size, their relationship with other interlocutors, the primary purpose of the conversation, and the context or modality in which it took place (e.g., face-to-face, online text, video call). This recalled experience served as a concrete grounding point for the subsequent discussion.

The semi-structured interview then proceeded, guided by 5 key focuses, with flexibility to ask follow-up questions and probe deeper into participant responses. These focuses, along with sample questions, included:

1. **Engagement in the Recalled Conversation:** We explored participants' subjective experiences within their recalled conversation to understand factors influencing engagement by asking them to recall key moments that they feel particularly engaged or disengaged. Sample questions were:
   - *"Thinking about that multiparty conversation, can you describe a moment when you felt a strong urge to interject? " — "What made that moment engaging for you?"*
   - *"Were there any moments where you felt disengaged or less involved?" — "What do you think contributed to that feeling?"*
2. **Reflection on Key Interactional Mechanisms:** Participants were prompted to reflect on how the interactions in their recalled multiparty conversation were managed, drawing on the types of mechanisms explored in Study 1 (though not necessarily using that academic terminology with them). The aim was to understand what aspects of conversational management they noticed and deemed important. Sample questions included:
   - *"How do you decide to take turns in the conversation?"* — "In your view, what is particularly effective or challenging?"
   - *"Thinking back to that conversation, did it mostly stick to one topic at a time, or were there often multiple topics or sub-conversations happening around the same time?" — "How did new topics usually get started in that group?" — "If there were multiple things being discussed, how did you personally keep track or decide which parts to focus on or respond to?"*
   - *"In that conversation, how could you tell if someone was actively engaged and paying attention, versus perhaps being distracted or less involved?" — "Were there any particular cues or behaviors that gave you a sense of others' level of participation or presence?" — "Were there times you consciously decided to be more 'visible' or active in the conversation? What made you do that?"*
   - *"How would you describe your own role in that conversation? What about others?" — "Did you find your role changing at different points? Did you perceive others' roles were changing?"*
3. **Hypothetical CA Integration:** We asked participants to imagine introducing a CA as a participant into their recalled conversation. Sample questions were:
   - *"Imagine you could have a CA join that specific multiparty conversation as another participant. If you were to replace one of the humans with this CA, who might that be, and what are your reasons?" — "What specific functions or roles would you want this CA to take on in that scenario? Or are there any roles you would definitely not want an AI to play in that context?"*

4. **Desirable CA Characteristics and Concerns:** This focus has shifted to broader expectations for CAs as social entities in multiparty settings. Our sample questions are:
   - *"Thinking more generally about a CA participating in a multiparty conversation much like a human participant, how would you feel about that? Would you accept it or even regard it as comparable to a human participant under certain conditions?"*
   - *"What favorable behaviors or social characteristics would you like to see? Please list as many as come to mind." — "What are your biggest concerns or reservations about having an AI as an active member in your multiparty conversations?"*
5. **Future Vision for CAs in Multiparty Conversations:** Finally, we prompted participants to envision ideal future scenarios by asking questions like:
   - *"If a CA could perfectly meet all your expectations for participating in a multiparty conversation, what would that look like?" — "Can you describe an ideal scenario with it?"*
   - *"Looking ahead, how do you envision CAs contributing to or enhancing your future multiparty conversations?"*

At the end of the interview, participants provided basic demographic information and reported their prior experience with CAs and LLMs. Each interview lasted approximately 40 minutes.

6.2. Participants

For Study 2, we recruited 15 participants who had not taken part in Study 1. Recruitment was conducted through a call for participation on social networking platforms (Table 2). Of these participants, 8 were based in the United States and 7 were based in China. The call invited individuals who expressed a deep interest in emerging conversational technologies to ensure thoughtful engagement with the interview topics. The recruited participants were 9 males and 6 females, with an average age of 28.3 years ($SD = 5.79$).

We also collected demographic information, including details about participants' prior experiences with conversational AI. All 15 participants reported having prior experience with CAs, such as text-based chatbots or voice-based assistants, with six indicating they were heavy users of these CAs. Furthermore, a substantial majority (14 out of 15 participants) reported experience with LLMs, and nine of these were frequent users of LLM-powered services. This widespread experience with contemporary AI tools ensured participants had a foundational understanding to draw upon when discussing their conceptualizations and expectations for CAs in multiparty interactions. The participant pool also included individuals with backgrounds in technology and AI/NLP research (N = 7), alongside participants from diverse professional fields. Each participant received compensation equivalent to approximately 8 USD in local currency.

6.3. Data collection and analysis

The interviews were audio-recorded and transcribed verbatim for analysis. To address RQ3, we employed thematic analysis, building upon the six-step framework by Braun and Clarke (Braun and Clarke, 2024) previously mentioned for Study 1. For Study 2, our analysis was primarily deductive, using the 5 predefined interview focuses (detailed in Section 6.1) as an initial structural framework for coding and interpreting the data.

First, the research team systematically reviewed the transcripts, and segments of participant responses were coded and initially categorized under the most relevant of the 5 interview focuses. Following this, we moved to a phase of theme development. This involved identifying patterns and initial thematic ideas *within each focus-coded set of data*. We then synthesized these findings by comparing concepts *across the different focuses*, merging similar ideas and integrating emergent insights to develop more comprehensive, overarching themes. For example, distinct points about CA roles arising from discussions in Focus 3 (Hypothetical CA Integration) and Focus 5 (Future Vision) might be consolidated into a singular theme on "Envisioned CA Roles".

7. Findings of study 2 - user perspective of CA in multiparty conversation

This section presents the findings from our semi-structured interviews conducted in Study 2, which explored user perspectives on integrating CAs into multiparty interactions (RQ3). The findings are organized into three main parts. First, we examine factors that participants identified as influencing their engagement in human multiparty settings, providing essential context for subsequent CA design considerations. Second, we delve into how participants envision the practical integration of CAs into multiparty conversations, including the potential roles CAs might assume and their positioning within the group. Finally, we outline the core persona traits that participants highlighted as crucial for CAs to be perceived as acceptable, trustworthy, and well-integrated participants.

7.1. Engagement in multiparty settings

Participants recalled recent multiparty conversations, identifying moments of engagement and disengagement, and reflected on the interaction strategies in those experiences. Analysis revealed two

Table 2
Demographics of semi-structured interview participants and information on recalled conversations.

Participant ID	Gender	Age	Conversation type*	Group size†	Interaction modality‡	Group familiarity§
P1	M	20s	task-oriented	3	online (text)	Acquaintances
P2	M	20s	social-oriented	4	online (text)	Acquaintances
P3	M	30s	social-oriented	4	online (text)	Acquaintances
P4	F	20s	social-oriented	3	face-to-face	Mixed
P5	F	10s	social-oriented	5	online (voice)	Acquaintances
P6	M	30s	task-oriented	12	online (text)	Acquaintances
P7	M	30s	task-oriented	3	face-to-face	Acquaintances
P8	M	20s	social-oriented	3	online (voice)	Acquaintances
P9	F	20s	social-oriented	4	online (text)	Acquaintances
P10	M	20s	social-oriented	3	online (text)	Mixed
P11	F	40s	task-oriented	4	face-to-face	Mixed
P12	F	20s	social-oriented	3	face-to-face	Strangers
P13	M	30s	social-oriented	4	face-to-face	Acquaintances
P14	F	20s	task-oriented	7	online (video)	Acquaintances
P15	M	20s	social-oriented	4	face-to-face	Acquaintances

* Conversation type describes the primary goal of the recalled conversation.
† Group size refers to the total count of individuals participating in the conversation.
‡ Interaction modality indicates the communication medium used for the conversation.
§ This column describes the familiarity of the interviewee with the other participants in the recalled conversation.

dominant criteria shaping these experiences: conversational inclusion and relevance alignment.

Conversational inclusion emerged as critical to engagement. Participants reported feeling engaged when conversation featured balanced participation, clear turn-taking, and equitable topic accessibility. As one participant noted:

> "When the conversations happen naturally, I don't have to consciously think about whether I should join in, or what should I say, this often marks the moment I feel engaged" *[P8]*

Conversely, disengagement arose when conversations became exclusionary, repetitive, or dominated by small subgroups. For example:

> "There was a time when a few people were constantly talking about things they were doing together, and since I didn't know the details, I felt a bit left out of the conversation." *[P5]*

Relevance alignment further influenced engagement. Participants felt invested when topics resonated with their personal interests or expertise. However, a strong focus on goal-driven agendas, particularly when such discussions lacked space for spontaneous and informal exchanges, was reported by participants as a cause of disengagement. One participant highlighted this tension:

> "Honestly, I never felt engaged (in this conversation). Even though it's my job, our work meetings are really dull. It's just the leader giving orders, and everyone else is responding out of obligation." *[P6]*

This finding aligns with insights from Study 1, where relevance-driven conversation was shown to motivate self-selected turns and guide the navigation of conversational threads.

Beyond inclusion and relevance, the disclosure of vulnerability played a nuanced role. Participants reported heightened engagement when conversations involved controlled vulnerability (e.g., sharing personal stories with deep feelings).

> "When one group member started openly sharing their deepest feelings and inner struggles, I felt fully engaged because it made me feel needed at that moment" *[P15]*

However, this positive effect of vulnerability was only realized when reciprocity was ensured. When such trust was breached, for instance, if disclosed vulnerability was exploited (e.g., by derailing a heartfelt story into gossip) or met with indifference, participants described a sharp decline in their engagement.

> "I was seriously seeking emotional support, but my friends thought I was joking, which made me feel a bit lonely... we end up chatting about trivial things." *[P2]*

Finally, excessive politeness in conversation was another factor participants reported as undermining engagement. They perceived such interactions as inauthentic and superficial, leading to disengagement.

> "We were all so 'nice' in that meeting. By the end, it felt like we'd talked for hours but decided nothing. I checked out mentally" *[P12]*

7.2. Envisioned agent integration

This subsection explores how participants conceptualize CA integration in multiparty conversations through their imagined role placements and future scenarios. By analyzing participants' responses to these interview prompts, we uncover their attitudes toward agents as conversational "parties" and their envisioned roles and party positioning.

7.2.1. Tool or peer?

Participants broadly expressed openness to integrating CAs as a "party" in multiparty interactions. However, their perspectives diverged sharply on whether CAs should occupy tool-like versus peer-like roles in these contexts.

A subset of participants maintained a utilitarian view, framing CAs as advanced tools rather than social equals—even if their capabilities become human-like.

> "even if they (CAs) become human-like, they are just moving from tools to servants." *[P6]*

This group emphasized functional boundaries, arguing that a CA's value lies in its ability to augment—not replace human agency in conversations.

In contrast, others adopted a more relational perspective, expressing willingness to accept CAs as equal participants if they contribute meaningfully.

> "I believe they can join conversation like any other human, as long as they bring something valuable to the conversation. if you can not tell (whether a certain party in conversation is an AI), then what's the matter?" *[P9]*

These participants framed CA's potential to be a social entity as contingent on the agent's ability to enhance conversation without disrupting social authenticity.

This tension between tool and social entity echoes longstanding debates in AI ethics about agency and anthropomorphism. Yet, what is clear is that both groups converged on the inevitability of AI integration in social or collaborative settings, regardless of its form or role.

7.2.2. What role and which "Party"

Participants envisioned CAs transcending current functional limitations, advocating for roles that adapt to dynamic social contexts rather than static task-oriented interactions. Their proposals emphasized two dimensions: role complexity and party positioning.

Participants argued that future CAs must move beyond today's narrow paradigms — such as information retrieval or meeting facilitation — to address nuanced, context-sensitive challenges. They described a broad spectrum of potential roles:

In educational settings, CAs in multiparty conversation could serve as peer learners, dynamically adjusting explanations based on learners' evolving comprehension levels. In familial settings, agents could act as generational mediators, helping to reframe parent–child conflicts with empathetic reasoning, or as bedtime storytellers, performing alongside parents to engage children in interactive narratives. In healthcare, CAs could function as group therapy supporters, offering an equal and private presence that detects subtle emotional shifts in patients' speech to assist therapists. Additionally, as medical information brokers, they could translate complex diagnoses into lay terms during family consultations, improving patient understanding. In specialized use cases, CAs could act as players in tabletop games when there aren't enough participants or serve as duet-like conversational partners, assisting memory-impaired individuals (e.g., Alzheimer's patients) in maintaining meaningful interactions with others.

These roles demand agents to operate with contextual awareness and social intentionality — capabilities participants felt current CAs lack.

When considering how CA should integrate into conversations, most participants envisioned agents as **extra parties**, augmenting human-human interactions without replacing existing roles.

However, a more novel perspective emerged: some participants imagined using a CA to **replace their own party**, effectively creating a digital doppelgänger that could represent them in meetings and social events on their behalf.

> "Sometimes I feel socially awkward, so I want to see how AI would handle multiparty conversations as me. So I can learn from it. In the meanwhile, this would let me engage in different social activities simultaneously. I think the only issue is how the AI me can keep my experiences in sync with me." *[P8]*

7.2.3. Concerns for CA integration

While many participants expressed openness and even enthusiasm for integrating CAs into multiparty conversations, this optimism was often tempered by significant concerns regarding the feasibility, appropriateness, and potential downsides of CAs acting as human-like participants. These concerns primarily centered on three aspects: privacy risks, accountability challenges, and the complexities of the CA's social integration and influence.

Privacy emerged as the most frequently mentioned concern. One facet of this was data security and the risk of collective surveillance, particularly if CAs were embedded in persistent group environments. Participants questioned data ownership and the potential for misuse, as P6 articulated:

> "If an AI is listening to team discussions, who owns that data? It feels like surveillance in disguise." *[P6]*

One more nuanced fear was cross-user information leakage. As CAs form familiarity with users over time, they necessarily collect extensive personal information for contextual understanding. Participants worried that a CA might inadvertently disclose private information shared by one individual to another during a multiparty conversation, or inappropriately carry information from one occasion to another. This potential for the CA to become an unintentional conduit for sensitive personal details was seen as a significant barrier to trust and adoption. P12 described this concern:

> "My biggest worry, if a CA is supposed to get to know all of us in a group over time, is that it might start mixing things up or letting secrets slip. Like, what if it casually mentions something about my weekend dating I only told it to my friend or even my colleague. That would be a disaster." *[P12]*

This also led to a second concern regarding unintended social consequences, particularly accountability.

The introduction of a CA as an active participant raised questions about where responsibility would lie if the CA made a mistake or provided incorrect information, especially in task-oriented contexts. P4 highlighted this responsibility gap:

> "If a conversational agent screws up a task, should the whole group take the responsibility?" *[P4]*

Such ambiguity in accountability was viewed as a potential source of conflict or inefficiency within groups relying on CA contributions.

A third area of significant concern centered on the CA's identity within the group and its potential impact on social dynamics. Participants questioned *who* or *what* the CA is: Is it an independent entity? An extension of the user who introduced it? Or does it represent some other outside interest? More importantly, who has the right to introduce a CA into a group interaction? As P6 noted, introducing a highly personalized CA into a group felt like a social imposition:

> "It's still like you bring your own friend to our conversation" *[P6]*

This might partly explain why many current CAs are designed as relatively neutral, "faceless" assistants, perhaps as a design compromise for collective acceptance.

Another aspect of CA's identity issue was raised by the need for the transparency of its non-human nature. Informing human participants that they are interacting with an AI, could significantly influence group dynamics compared to an all-human setting. For instance, P7 considered a scenario where a CA is integrated in multiparty conversation as his boss:

> "If the CA starts giving advice, how seriously do I take it? Will I feel more free to express my disagreement, or will I just withhold my thoughts since I know it is just an AI bot?" *[P7]*

Considering all, addressing these multifaceted concerns is clearly a prerequisite for successful CA integration.

7.3. The ideal persona: traits of a socially-integrated agent

This subsection synthesizes participants' visions for an ideal persona-driven CA in multiparty conversations. The concept of a persona, a cohesive and recognizable character, is crucial, as users naturally anthropomorphize CAs and expect them to behave like social actors rather than mere tools (Nass et al., 1994; Sun et al., 2024). Drawing from participants' reflections and their expectations for social behaviors, we derive four core traits that constitute this ideal persona. These traits collectively define a character that balances functional utility with sophisticated social intelligence, enabling the CA to act and be perceived as a genuine participant.

7.3.1. Being proactive

For a CA to embody the persona of a genuine participant, proactivity is a foundational trait — an "admission ticket" to the conversation, as participants indicated. When envisioning this ideal persona, participants often contrasted it with the limitations of existing CAs they were familiar with.

Many noted that current AI-based CAs, either passively respond to human input (e.g. Siri, ChatGPT), or intrusively take the turn to give suggestions(e.g. GitHub Copilot). This kind of reactive or overly simplistic initiative model is misaligned with the persona of an intelligent collaborator suitable for multiparty conversation.

> "Current chatbots and voice assistants clearly can't participate in group conversations. I think they need to be able to make autonomous decisions." *[P5]*

Participants described that ideal CAs must be fundamentally proactive in how they participate in conversation. This proactive trait manifests in behaviors that project the character of a socially aware and intelligent collaborator. These behaviors include: dynamically sharing the initiative with human users, understanding conversational context deeply, predicting other participants' intentions and potential conversational trajectories, identifying opportune and relevant moments to contribute, and timing their interventions appropriately with social awareness.

More deeply, this proactivity is not just about starting contributions but also about judiciously managing them. Beyond initiating new conversational turns, threads, or topics, participants stressed that it is equally important for CAs to know when and how to actively step back from their own contributions to avoid dominance.

> "Starting a topic is easy. Knowing when to shut up? That's the hard part." *[P1]*

Furthermore, envisioned proactive behaviors extended beyond discrete speaking turns. Participants suggested that a CA's persona should not feel constrained to a rigid turn-based model. Instead, CAs should be able to interrupt reasonably when necessary, provide timely feedback, and importantly, use backchannels effectively to project a persona of active listening and continuous engagement. This use of backchannels aligns with the presence management strategies observed in human conversations in Study 1 and could help innovate CA interaction from a strictly turn-based model to a more continuous paradigm, as echoed in prior HCI research (van Berkel et al., 2021).

> "I need it to really listen to me and the other people in the group, instead of every time it intervened, make the conversation a monologue and splitting group conversation as multiple human-ai 1on1 chats...they should also be able to point out mistakes and act as a respectful and smart assistant." *[P8]*

7.3.2. Holding long-term relationships

A key trait distinguishing a meaningful persona from a generic tool is the capacity for continuity. As human interactions deepen over time, people naturally form long-term bonds, and this development of long-term relationships decreases interaction friction. Participants perceived

current CAs as lacking this capacity for continuity, projecting a persona of a transient stranger who treats each interaction as a discrete event. This limitation, they felt, prevents CAs from being accepted as a stable member of a group.

> "Every time I talk to my ChatGPT, it's like meeting a new intern — polite but clueless about yesterday's work." *[P10]*

Participants emphasized the importance of agents being able to recall past interactions, recognize familiar individuals, and keep tracking participants' personal experiences in multiparty conversation. Otherwise, as P6 noted, *"it is like you invite a new stranger to multiparty conversation"*.

Without these abilities, participants expressed their concerns that CAs could become a burden in multiparty conversations, disrupting established social dynamics and requiring extra effort from human participants to constantly fill the CA's contextual information gap.

This personality trait is functionally enabled by a long-term memory system.

Although many dyadic AI agents, such as the latest ChatGPT, have already been equipped with memory systems, this memory only updates after direct interactions with the agent. Participants envisioned a persona whose long-term memory could be automatically updated with context about group members, regardless of whether the CA actively participates or not. This perception of an ever-aware persona was also seen as crucial for building trust.

> "The AI should learn from our chats even when it's silent, or at least ask questions on its own. Otherwise, when we're working on tasks together, it'd be a hassle to keep updating things to it." *[P4]*

Participants suggested long-term relationships could be built in two main scenarios: 1) attaching the CA to a specific environment (e.g., a meeting room, a family home), or 2) associating the CA with certain people as a personalized conversational companion. These models reflect a desire for CAs that are either deeply embedded within a consistent context or closely tied to individual users to foster continuity and understanding over time.

7.3.3. Exhibiting social authenticity

Participants emphasized that for CAs to feel like genuine conversational partners, they must exhibit social authenticity, where the CA's presence and contributions feel real and natural within the conversational flow. This multifaceted trait involves the CA's ability to dynamically manage its own presence, engage in flexible and contextually appropriate chit-chat, and maintain a consistent and believable character.

Much like how humans manage their presence in multiparty conversations, agents should be able to transition between being "visible" and "invisible" as needed.

> "I want it to feel like a real person. Maybe give me a little nudge now and then to show it's there, but I don't want it to be too overbearing on showing its presence" *[P5]*

Being flexible in becoming invisible refers to the agent's ability to adjust its presence and role based on the context. The agent should be able to recognize moments when its involvement is unnecessary. It can be visible in certain conversation threads while remaining invisible in others, seamlessly blending into the dynamic nature of multiparty conversations.

Secondly, participants highlighted the importance of flexible, human-like chit-chat for social authenticity, especially in task-oriented contexts. This kind of small talk could make a CA embody a persona that feels more like an integrated group member and less like a purely functional tool, mirroring how humans build rapport (Sun et al., 2021). As P11 suggested, a simple, contextually relevant social comment could make a significant difference:

> "If we're planning a project, I would like to see a quick *'How's your dog'*, which can make the AI feel human—not just a taskmaster." *[P11]*

Finally, to achieve social authenticity, participants noted that a CA should develop and maintain a consistent identity or persona with its own experiences or 'history'. They envisioned the CA having its own consistent interests, preferences, knowledge, or even personal experiences, rather than shifting these often to flatter the user.

> "I hope this AI can have its own virtual life, like having a favorite football team. But it shouldn't just flatter the user by changing its preferences all the time, like, liking football today and basketball tomorrow. It should keep its experiences and memories consistent and continue to develop..." *[P8]*

Participants believed that such consistency and gradual evolution would significantly enhance a CA's perceived social authenticity, helping it establish a more believable and well-defined character, thereby fostering deeper engagement.

7.3.4. Showing imperfections

Counter-intuitively, a final critical trait for a believable persona is a degree of human-like imperfection. This was not a desire for flawed or unreliable CAs, but rather a rejection of overly polished, unnaturally "perfect" agents whose personas feel inauthentic. Participants indicated that such imperfections, when appropriately displayed, could enhance a CA's genuineness and relatability across two main dimensions: its social characteristics and its perceived abilities.

First, regarding social characteristics, participants desired a persona capable of more than just basic politeness, envisioning agents with nuanced emotional intelligence, empathy, cultural sensitivity, and humor. For instance, they described true emotional intelligence not simply as using expressive language, but as a CA's capacity to adapt its emotional expressions appropriately to specific group dynamics and interactional contexts. This might involve showing controlled anger with an uncooperative participant while being gentle and supportive with others, a nuance often missing in current CAs, as P14 noted:

> "Current LLMs are human value aligned too much... too polite to be useful, in group, we need a certain of aggression" *[P14]*

Participants complained that current LLM-based CAs are often fine-tuned to be overly positive or excessively polite, resulting in a "flattering" impression. This can distort user expectations, limit opportunities for critical feedback, and ultimately diminish trust in these agents. For multiparty conversations, participants believed a CA's persona should also incorporate the ability to deliver appropriately critical or negative responses when the situation calls for it, thereby fostering more open and honest interactions.

Second, participants valued imperfections related to a CA's perceived knowledge and abilities, emphasizing that a CA should not be presented as omniscient or infallible. A key aspect of this was the desire for CAs to acknowledge their own limitations or "insufficient ability", as P3 described:

> "I wish the agent to acknowledge its insufficient ability. Although this kind of 'honesty' is actually faked, it makes AI feel more real to me. I'd see it as a cooperative coworker... If AI assists us in every aspect, I see it could make group conversations more closed off and decrease collaboration." *[P3]*

An agent that can express *"Hey Alice, I cannot do that; Bob is better than me on that."* is more socially accepted and better for collaboration and other human-centered interactions.

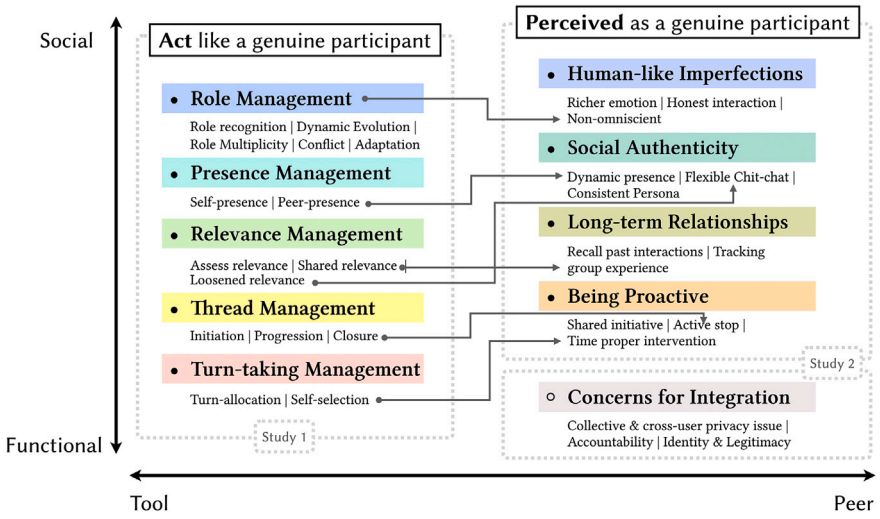


Fig. 4. A Conceptual Model for CA in Multiparty Conversation as Genuine Participant, Synthesizing Human Interaction Mechanisms (Study 1) and User Expectations (Study 2).

8. Designing conversational agents for genuine multiparty participation

Synthesizing the findings from our two studies, this section first introduces a conceptual model for designing CAs that are capable, believable, and acceptable in multiparty conversations. We then derive a set of design considerations from this model to guide future multiparty CA development.

8.1. Bridging human interaction and user expectation

Building upon the complementary findings from our two studies, we propose a conceptual model for designing CAs capable of genuine participation in multiparty conversations (Fig. 4). The model is structured to provide a clear roadmap for designers by synthesizing two critical dimensions: the functional mechanics of how a CA must act (derived from Study 1) and the essential traits a CA should embody(derived from Study 2).

The left column of the model, grounded in the findings of Study 1, details the foundational capabilities required for a CA to act like a genuine participant. It outlines the 5 core interactional mechanisms that humans adeptly employ to navigate multiparty conversations: Turn-taking, Thread, Relevance, Presence, and Role Management. These are arranged along the vertical axis from more functional skills (e.g., managing discrete turns) to more holistically social skills (e.g., adapting complex social roles). Mastering these mechanisms is the technical and behavioral prerequisite for any CA aiming for effective participation. It is the core repertoire of actions the CA must be able to perform.

The right column of the model addresses the crucial second dimension: the essential qualities a CA must embody to be perceived as a genuine participant. As Study 2 revealed, a well socially integrated CA is contingent on its possessing a desirable persona. This column outlines the 4 core traits that users envision: Being Proactive, Holding Long-term Relationships, Showing Social Authenticity, and Showing Human-like Imperfections. This dimension also acknowledges the concerns for multiparty CA integration, which act as critical trust factors that must be managed for the persona to be accepted. These traits define the CA's character and determine whether it can successfully move beyond the role of a "Tool".

The model also illustrates the direct connections between the findings from our two studies, showing how the interactional mechanics (left) enable the desired persona traits (right). For instance, to successfully exhibit the trait of Being Proactive, a CA must draw upon the mechanisms of Turn-taking Management (to self-select turns) and Thread Management (to actively initiate and close topics). Similarly, Relevance Management is a versatile mechanism that underpins multiple traits. Its "loosened relevance" facet allows for the chit-chat essential for showing social authenticity, while its ability to track shared relevance over time is the foundation for maintaining long-term relationships.

By framing design in this way, the model offers a concrete path forward. Designers can begin by implementing the core interactional mechanics and then treat these as a palette for tuning specific, socially intelligent traits. The goal of this process is to develop CAs that are not only capable in their function, but also perceived as believable, acceptable, and socially well-integrated by their human partners.

8.2. Design considerations

8.2.1. Mandate proactivity as a foundational capability

Both studies highlighted the dynamism of human multiparty conversations, with Study 1 showing extensive use of self-selection in turn-taking and proactive strategies for managing presence and roles. Echoing this, Study 2 underscored that users expect CAs to be "fundamentally proactive" — viewing this as an "admission ticket" to genuine participation, contrasting with their perception of current CAs as often too passive or having overly simplistic initiative models.

This is more than simply enabling CAs to take turns; it requires designing them to make autonomous decisions grounded in a deep contextual understanding of the ongoing multiparty setting. Specifically, proactive CAs should be able to dynamically share conversational initiative with human users, identifying opportune moments to contribute, judiciously withhold contributions, and actively signal continuous engagement, for example, by using backchannels.

To achieve this level of sophisticated proactivity, future CA designs should incorporate and advance the reasoning and planning capabilities inherent in LLMs. Existing research on LLM-based multiparty CAs, such as the Swimmy (Liu et al., 2025) and the Koala (Houde et al., 2025) demonstrates promising directions in utilizing such capabilities. However, we acknowledge that individuals may have varying preferences for proactivity depending on the context. Therefore, developing more context-dependent proactivity parameters and providing users with clear mechanisms to customize the CA's level of initiative will be crucial areas for future design and research. For instance, rather than a generic frequency setting, systems could provide granular, explainable parameters such as "critical level," which determines intervention necessity through critical reasoning, or "experience focus," which regulates contributions based on the agent's accumulated memories.

8.2.2. Enable conversational flexibility

Human multiparty conversations are not rigidly structured but are dynamic and open-ended, requiring participants to be highly flexible. Study 1 demonstrated this through participants' adept use of "loosened relevance" to navigate social goals alongside task objectives and their fluid "role adaptation" in response to the evolving conversational landscape. Reinforcing this, Study 2 highlighted users' desire for CAs that exhibit similar adaptability, such as through "dynamic presence", which means the CA being able to be visible or invisible appropriately, and the ability to engage in contextually relevant "chit-chat" to enhance social authenticity. These findings suggest that for CAs to be truly integrated, they must mirror some of this human capacity for nuanced, flexible interaction.

Current CAs often lack this crucial flexibility. They tend to enforce rigid adherence to predefined conversational paths or relevance criteria, exhibit limited and often artificial presence management, and are typically confined to static roles. This rigidity often causes them to disrupt the fluid flow of multiparty conversation.

Achieving such conversational flexibility demands that CAs grasp implicit social norms, read subtle contextual cues, and balance task-focused input with socio-emotional support. Future work should develop models that learn these nuances and embed an ongoing self-reflection loop so agents can adjust their behavior in real time.

8.2.3. Design a consistent and believable CA Persona

User expectations from Study 2 highlight a clear vision for a persona-driven CA, defined by a consistent identity that can evolve its knowledge and understanding of context over time. This persona should comprise distinct interests, habits, and background knowledge, which together enhance its perceived social authenticity and establish it as a more realistic, non-omniscient participant. Participants identified these traits as foundational for building user trust and enabling more meaningful long-term relationships. Furthermore, a coherent persona, with its established knowledge base and "experiences", can empower the CA to be more effectively proactive. It provides a stable internal framework for assessing relevance, deciding when to contribute, and managing its presence. For example, when the conversation falls outside its interests or expertise, the CA chooses to listen.

In contrast, many current CAs lack this continuity of character or a clear persona, often interacting generically without a stable set of internal states. Designing such a persona requires advanced memory architectures and mechanisms for consistent knowledge integration and retrieval. Designers must also carefully consider how this persona is established, whether predefined or learned. While the former ensures control and consistency, the latter offers adaptability and personalization but carries risks of unpredictability. Alternatively, adopting a hybrid approach may help to ensure its evolution remains coherent, enhances the user experience, and avoids unintended biases.

8.2.4. Broaden evaluation metrics

Our findings across both studies demonstrate that genuine participation in multiparty conversations is a complex, multifaceted process, involving much more than conversation goal achievement. Current evaluation practices for CAs, however, often prioritize task-specific metrics (e.g., dialogue efficiency, task success rates) or general measures of perceived anthropomorphism and usability. While these are important, they may not adequately capture how well a CA integrates into the social fabric of a group or how effectively it manages the core mechanisms of human conversation.

To truly assess a CA's ability to act as a genuine participant, evaluation metrics and methods must therefore evolve. Designers and researchers should consider evaluating CA performance on various dimensions of the interactional mechanisms identified in Study 1. For example, this could involve asking users to rate or describe how well the CA self-selects turns, how coherently it navigates conversational threads, or how dynamically and appropriately it signals its presence.

Future work in this area should focus on developing more comprehensive evaluation frameworks and creating new metrics specifically tailored for multiparty human-AI interaction. This includes the challenge of operationalizing complex user-desired qualities, such as "social authenticity" or "appropriate proactivity", into measurable or reliably assessable constructs, and designing study protocols that can effectively capture the intricate interplay within group settings.

9. Discussion

In this section, we discuss two of the most significant themes that emerged. First, we explore the critical role of agency and autonomy in designing CAs for multiparty contexts. Second, we reflect on the concept of "humanness," questioning what it truly means for an agent to be a genuine participant.

9.1. Toward agency and autonomy design in multiparty conversation

Our findings reveal 5 essential mechanisms underpinning successful multiparty conversations, including turn-taking, thread, relevance, presence, and role management. All these mechanisms are fundamentally grounded in human agency and autonomy, highlighting participants' ability to act with self-causality and independence, which ensures meaningful contributions to the conversation.

Interestingly, the conversational "agent" actually embodies the vision of the "agency", with the implicit design goal of *"to perform by itself"*. Considering user expectations, it is about designing agency and autonomy in future conversational agent systems. In interpersonal interaction, a participant with sense of agency and autonomy is more likely to be intrinsically motivated, leading to increased engagement, and fostering feelings of trust and connectedness (Bennett et al., 2023). A similar effect may apply to human-AI relationships; however, this also raises some concerns. For instance, how will people interact with a "servant" that has too much autonomy? Or, when the agent acts as a digital doppelgänger, will people really hand over their agency, and who should take the responsibility?

As advocated by Calvo et al., designing for agency and autonomy is not just a functional need, but an ethical imperative (Calvo et al., 2020). Therefore, designing for agency in CAs should be framed as supporting, not supplanting, human agency. Ensuring that CAs empower and engage users, rather than diminish their presence, is important. Our second study reinforces this perspective, as participants expressed no desire for omniscient CAs.

As agency and autonomy have always been fundamental concepts in HCI (Bennett et al., 2023), this suggests multiparty CA design should follow a scaffolding agenda: initially supporting functional needs (current state), then developing agents with the autonomy to engage in human interactions, and ultimately designing agents that enhance human agency. Future research should explore how CAs can effectively balance machine agency and human agency, fostering healthier human-AI collaboration.

9.2. Rethinking the humanness of conversational agent

Human-likeness or anthropomorphism has long been a central focus in the research on CAs. In our study, participants expressed a desire for agents to feel socially human-like, including qualities such as a dynamic social presence, consistent personal experiences, and imperfect social traits. While some studies suggest that excessive human-likeness can trigger negative reactions (Kim et al., 2022), most demonstrate that perceived humanness positively influences users' overall evaluation of CAs (Shalini Chandra and Srivastava, 2022).

However, this raises a fundamental question: should CAs internally function like humans — mimicking human cognitive processes — or is it sufficient for them to merely be perceived as human-like? For instance, in turn-taking, a truly human-like approach would involve an agent deciding to speak based on internal relevance and intent. However, an agent could instead select moments to intervene based on statistical patterns and still be perceived as anthropomorphic, especially in casual

conversations with loosened relevance. While designing for perceived human-likeness is more practical, the nuance this distinction casts on the long-term user experience remains an open question.

This dilemma echoes the philosophical debates about intelligence since the inception of the "Turing Test" (Turing, 2009). It also aligns with ongoing discussions about whether LLMs can be World Models (Ha and Schmidhuber, 2018). Notably, insights from this discussion suggest that grounding language in real-world perception and experience is beneficial for LLM performance (Peng et al., 2024). We envision that deeper exploration of human-likeness in CAs may not only refine the design of AI systems but also inspire broader inquiries into the nature of human intelligence itself. By understanding and rethinking what it means for AI to appear and function like humans, we can develop more sophisticated and natural human-AI interactions.

10. Limitations

There are several limitations in our study that should be acknowledged.

First, the participants in our study are mainly from the U.S. and China, which may introduce cultural bias in the findings. While we aimed to capture diverse perspectives within these contexts, the cross-cultural aspects of multiparty conversations remain underexplored in this work. Cultural differences can significantly affect conversational norms and social behaviors (Singelis and Brown, 2006), such as turn-taking conventions (e.g., tolerance for overlap or silence), directness in communication, expressions of politeness, or the perceived appropriateness of "loosened relevance". Consequently, some details of the 5 mechanisms and the prioritization of desirable CA traits might differ in cultural settings. Further studies with more diverse cultural representations are necessary to enhance the applicability of our findings.

Second, the conversational scenarios in our study (Topics 1–4), whether involving casual social exchanges or task-oriented discussions, were predominantly cooperative and functional. We acknowledge that hostile, adversarial, or highly competitive multiparty interactions may involve different mechanisms or different priorities that were not captured in this work.

Regarding the latter, the group size in Study 1 was constrained to 3 participants. While this controlled setup allowed us to capture core multiparty dynamics, it may not fully reflect the complex coordination processes inherent in larger groups. For instance, in larger groups, the number of active conversational threads could increase significantly, which might lead participants to adopt different or additional thread management strategies. Future research should therefore investigate how the mechanisms we identified scale and evolve in larger, more complex social settings.

Finally, our study's methodological design sets important boundaries for interpreting the findings. Study 1's conversations were limited to 5 minutes, a constraint that directed our analysis toward synchronous, text-based interactions. This design prevented the observation of dynamics that unfold over longer time spans or in asynchronous communication, such as trust-building or the revival of dormant threads. The imposed time limit may also have introduced an artificial sense of urgency, simplifying conversational structures. Together with the uncontrolled contextual variables in Study 2's recalled experiences, these constraints mean that several contexts — multimodal conversations, or interactions among less familiar participants — remain only sparsely represented in our dataset.

As a result, our findings and discussion primarily concern the design of CAs for synchronous, text-based multiparty conversations, which we expect to remain a dominant mode of human–AI interaction in the near future. At the same time, we regard these sparsely represented contexts as highly valuable avenues for future research. Extending our framework to longer durations, asynchronous exchanges, and multimodal settings will be essential for validating and broadening the applicability of the human-centric insights identified here.

11. Conclusion

Multiparty conversation is an integral part of daily life, yet enabling CAs to participate naturally in such interactions remains a significant challenge. This study offers a comprehensive qualitative exploration of the intricate dynamics of multiparty conversations from a human-centric perspective, deepening our understanding of how people interact in these settings and what they expect from CAs designed to engage alongside them.

Our research yields two primary sets of insights. Study 1 identified 5 key mechanisms that enable humans to genuinely participate in text-based multiparty conversations: turn-taking, threading, relevance, presence, and role management. Study 2 then defined an ideal persona that users envision for a CA, which is constituted by 4 traits: being proactive, holding long-term relationships, showing social authenticity, and exhibiting human-like imperfections. By synthesizing these findings, this paper outlines a conceptual model that links human interactional mechanics to the formation of this ideal CA persona. From this model, we distill a set of design considerations to guide the development of future systems that are capable, believable, acceptable, and socially well-integrated in group settings.

CRediT authorship contribution statement

Shitao Fang: Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **Xingyu Liu:** Writing – review & editing, Formal analysis, Data curation, Conceptualization. **Takeo Igarashi:** Writing – review & editing, Supervision, Resources, Funding acquisition. **Koji Yatani:** Writing – review & editing, Supervision, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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